

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.365798034	delta=+0.0142004
ratio=1.0060386		
CoG x [m]	-0.002024708562 -> -0.002367101188	delta=-0.000342393
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.0008738491313	delta=+0.000904376
ratio=-		
CoG z [m]	0.009509749599 -> 0.01052070667	delta=+0.00101096
ratio=-		
rotor effectiveness 1	1 -> 0.9653611597	delta=-0.0346388 ratio=0.96536116
rotor effectiveness 2	1 -> 0.9732088814	delta=-0.0267911 ratio=0.97320888
rotor effectiveness 3	1 -> 0.9582561006	delta=-0.0417439 ratio=0.9582561
rotor effectiveness 4	1 -> 0.9549408752	delta=-0.0450591 ratio=0.95494088
rotor command lag [s]	0.004997730255 -> 0.2	delta=+0.195002 ratio=-

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06501125892, -2.509027419e-05, 2.25800274e-05]
[-2.509027419e-05, 0.06525006997, 9.368878049e-06]
[ 2.25800274e-05, 9.368878049e-06, 0.1289941839]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06500864275	delta=+5.59976e-05
ratio=1.0008621		
principal moment 2 [kg m^2]	0.06500006701 -> 0.0652526768	delta=+0.00025261
ratio=1.0038863		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289941932	delta=+2.07688e-06
ratio=1.0000161		

Optimization

joint dynamics loss: 1.07920472774
Gaussian-prior cost: 0.336426434718
selected strict-ZOH rotor lag [s]: 0.2

Bag failure_1_single

reconstruction position RMSE [m]: 0.000652642995943
reconstruction position component RMSE [m]: [0.00034 0.000324 0.000453]
reconstruction orientation RMSE [deg]: 0.596167087291
gyro consistency RMSE xyz [rad/s]: [0.041076 0.085964 0.026575]
specific-force consistency RMSE xyz [m/s^2]: [0.938568 0.163511 0.339544]
inferred external force RMS xyz [N]: [0.785908 0.385405 2.995019]
inferred external torque RMS xyz [N m]: [0.072144 0.242564 0.106157]
bag dynamics loss: 1.07920472774
gimbal command lag [s] 0.004998683929 -> 0.04021704971

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message=`gtol` termination condition is satisfied.
reported optimality=1.8594785267112002e-08
initial cost=1.4156315076964658
final cost=1.4156311624620588
cost reduction=3.452344070620228e-07
coordinate step L2=0.0005127896673439322
final gradient inf-norm=1.8594785267112002e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=406.9245366549322
max directional finite-difference Jacobian relative error=7.904933618886317e-12